

ASSIGNMENT NO.I

GAMAT301: MATHEMATICS FOR COMPUTER AND INFORMATION SCIENCE-3

To be submitted at Room No. 302 on my table before 3 PM Wednesday, 23-07-2025.

1. The length of time (in minutes) that a lady speaks on the telephone is found to be random phenomenon, with a probability function given by

$$f(x) = Ae^{-x/5} \text{ for } x \geq 0$$

$$= 0, \text{ otherwise}$$

- (i) Find the value of A such that $f(x)$ is a pdf
- (ii) find the distribution function of X
- (iii) What is the probability that the number of minutes the lady talks over the phone is
 - (a) More than 10 minutes
 - (b) Less than 5 minutes
 - (c) Between 5 and 10 minutes
- (iv) Find the mean and variance of X .

2. For the discrete probability distribution

x	: 0	1	2	3	4	5	6	7
$f(x)$	: 0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

Determine the following:

- (i) The value of K
- (ii) CDF
- (iii) $P(X < 6)$ (iv) $P(X \geq 6)$ (v) $P(0 < X < 5)$ (vi) mean (vi) variance

3. A random variable X has the CDF

$$F_X(x) = 0, \quad x < -1$$

$$= A(1 + x), \quad -1 \leq x < 1$$

$$= 1, \quad x \geq 1$$

- (a) What is the value of A ?
- (b) With the above value of A , what is $P[X > 1/4]$?
- (c) With the above value of A , what is $P[-0.5 \leq X \leq 0.5]$?
- (d) find $P[-2 < x < 0]$ (e) mean and variance

4. In a game, a die is thrown. If the throw results in an even number you win Rs.2. If the outcome is 1 or 3, you lose Rs.1 and if the outcome is 5, you lose Rs.3. Find the expected amount of winning and its S.D. Write down its distribution function.

5. If $f(x) = 2x$, $0 \leq x \leq b$, find b such that it is a valid probability density function. Also find mean, S.D. and CDF.